

Unit 4

Higher Order Linear Equations:

Theory of Higher Order Linear Equations

The n^{th} order linear equation is in the form

$$P_0(t)y^{(n)} + P_1(t)y^{(n-1)} + \dots + P_{n-1}(t)y' + P_0(t)y = G(t) \quad \dots \quad (1)$$

where $P_0, P_1, P_2, \dots, P_n$ and G are continuous real valued functions in an open interval $I = (\alpha, \beta)$

and $P_0(t) \neq 0$ for all $t \in I$. Dividing (1) by $P_0(t)$, we get

$$L[y] = y^{(n)} + p_1(t)y^{(n-1)} + \dots + p_{n-1}(t)y' + p_n(t)y = g(t) \quad \dots \quad (2)$$

Equation (2) can be assigned the initial conditions

$$y(t_0) = y_0, y'(t_0) = y'_0, \dots, y^{(n-1)}(t_0) = y_0^{(n-1)} \quad \dots \quad (3)$$

where $t_0 \in I = (\alpha, \beta)$ and $y_0, y'_0, \dots, y_0^{(n-1)}$ are given real numbers to form higher order Initial Value Problem (IVP).

Theorem 1 Let $p_1, p_2, \dots, p_n$ and g be continuous in an open interval I . Then, there exists exactly one solution $y = \phi(t)$ of the differential equation $y^{(n)} + p_1(t)y^{(n-1)} + \dots + p_{n-1}(t)y' + p_n(t)y = g(t)$ with initial conditions

$$y(t_0) = y_0, y'(t_0) = y'_0, \dots, y^{(n-1)}(t_0) = y_0^{(n-1)}.$$

This solution exists throughout the interval I .

n^{th} Order Homogeneous Equation

The n^{th} order homogeneous equation is

$$L[y] = \frac{d^n y}{dt^n} + p_1(t) \frac{d^{n-1} y}{dt^{n-1}} + \dots + p_{n-1}(t) \frac{dy}{dt} + p_n(t)y = 0$$

That is,

$$L[y] = y^{(n)} + p_1 y^{(n-1)} + p_2 y^{(n-2)} + \dots + p_{n-1} y' + p_n y = 0 \quad \dots \quad (1)$$

If $y_1, y_2, \dots, y_n$ are solutions of (1), then the linear combination

$$y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n \quad \dots \quad (2)$$

is also a solution of (1), where $c_1, c_2, \dots, c_n$ are arbitrary constants.

The values of $c_1, c_2, \dots, c_n$ can be determined by the initial conditions

$$y(t_0) = y_0, y'(t_0) = y'_0, \dots, y^{(n-1)}(t_0) = y_0^{(n-1)}.$$

i.e.

$$\begin{aligned} c_1 y_1(t_0) + c_2 y_2(t_0) + \dots + c_n y_n(t_0) &= y_0 \\ c_1 y'_1(t_0) + c_2 y'_2(t_0) + \dots + c_n y'_n(t_0) &= y'_0 \\ \dots &\dots \dots \dots \dots \dots \dots \dots \dots \dots \\ c_1 y_1^{(n)}(t_0) + c_2 y_2^{(n)}(t_0) + \dots + c_n y_n^{(n)}(t_0) &= y_0^{(n)} \end{aligned}$$

The Wronskian of $y_1, y_2, \dots, y_n$ is

$$W(y_1, y_2, \dots, y_n) = \begin{bmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & y_2' & \dots & y_n' \\ \dots & \dots & \dots & \dots \\ y_1^{(n-1)} & y_2^{(n-2)} & \dots & y_n^{(n-1)} \end{bmatrix}$$

The equation (1) will have a unique solution if $W(y_1, y_2, \dots, y_n) \neq 0$ and will have no solution if $W(y_1, y_2, \dots, y_n) = 0$.

If equation (1) has a unique solution, then

$$\begin{aligned} c_1 &= \frac{1}{W} \begin{bmatrix} y_0 & y_1 & \dots & y_n \\ y_0' & y_1' & \dots & y_n' \\ \dots & \dots & \dots & \dots \\ y_0^{(n-1)} & y_1^{(n-2)} & \dots & y_n^{(n-1)} \end{bmatrix} \\ c_2 &= \frac{1}{W} \begin{bmatrix} y_1 & y_0 & \dots & y_n \\ y_1' & y_0' & \dots & y_n' \\ \dots & \dots & \dots & \dots \\ y_1^{(n-1)} & y_0^{(n-2)} & \dots & y_n^{(n-1)} \end{bmatrix} \\ &\dots \dots \dots \dots \dots \dots \dots \dots \\ &\dots \dots \dots \dots \dots \dots \dots \dots \\ c_n &= \frac{1}{W} \begin{bmatrix} y_1 & y_2 & \dots & y_0 \\ y_1' & y_2' & \dots & y_0' \\ \dots & \dots & \dots & \dots \\ y_1^{(n-1)} & y_2^{(n-2)} & \dots & y_0^{(n-1)} \end{bmatrix}. \end{aligned}$$

Example 1 Find the interval in which the solution of the following differential equation is certain to exist.

- a) $ty''' + \sin ty'' + 3y = \cos t$
b) $t(t-1)y^{(4)} + e^t y'' + 4t^2 y = 0$

Solution

The given differential equation is

$$\begin{aligned} t(t-1)y^{(4)} + e^t y'' + 4t^2 y &= 0 \\ y^{(4)} + \frac{e^t}{t(t-1)} y'' + \frac{4t}{(t-1)} y &= 0 \end{aligned}$$

Here, $p_1(t) = \frac{e^t}{t(t-1)}$, $p_2(t) = \frac{4t}{(t-1)}$

Since $p_1(t)$ and $p_2(t)$ are continuous for all t except $t = 0$ and $t = 1$

$\therefore$ the solution of (1) lies in the interval

$$-\infty < t < 0, 0 < t < 1, 1 < t < \infty$$

Example 2 Verify that the functions e^t , e^{-t} , e^{-2t} are solutions of the differential equation $y''' + 2y'' + y' - 2y = 0$ and determine their Wronsteian.

Solution:

The given differential equation is

$$y''' + 2y'' + y' - 2y = 0 \quad \dots \quad (1)$$

Let $y_1(t) = e^t$, $y_2(t) = e^{-t}$, $y_3(t) = e^{-2t}$ and so,

$$\begin{aligned} y_1'(t) &= e^t, y_2'(t) = -e^{-t}, y_3'(t) = -2e^{-2t} \\ y_1''(t) &= e^t, y_2''(t) = e^{-t}, y_3''(t) = 4e^{-2t} \end{aligned}$$

$$y'''_1(t) = e^t, y'''_2(t) = -e^{-t}, y'''_3(t) = -8e^{-2t}$$

Substituting these values in (1), we get

$$e^t + 2e^t - e^t - 2e^t = 0 \\ \Rightarrow 0 = 0, \text{ true.}$$

$\therefore e^t$ is a solution of (1)

Similarly, $y_2(t)$ and $y_3(t)$ are also solution of (1)

$$\begin{aligned} \text{Again, } W(y_1, y_2, y_3) &= \begin{vmatrix} y_1 & y_2 & y_3 \\ y'_1 & y'_2 & y'_3 \\ y''_1 & y''_2 & y''_3 \end{vmatrix} = \begin{vmatrix} e^t & e^{-t} & e^{-2t} \\ e^t & -e^{-t} & -2e^{-2t} \\ e^t & e^{-t} & 4e^{-2t} \end{vmatrix} \\ &= e^t \cdot e^{-t} \cdot e^{-2t} \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & -2 \\ 1 & 1 & 4 \end{vmatrix} = e^{-2t} \begin{vmatrix} 0 & 0 & 1 \\ 2 & 1 & -2 \\ 0 & -3 & 4 \end{vmatrix} = e^{-2t} \begin{vmatrix} 2 & 1 \\ 0 & -3 \end{vmatrix} \\ &= e^{-2t}(-6 - 0) = -6e^{-2t} \neq 0. \end{aligned}$$

Since $W \neq 0$, the equation (1) has a unique solution.

Theorem 2 Let the functions $p_1, p_2, \dots, p_n$ be continuous in an open interval I and let the functions $y_1, y_2, \dots, y_n$ be solutions of

$$y^{(n)} + p_1(t)y^{(n-1)} + \dots + p_{n-1}(t)y' + p_n(t)y = 0 \quad \dots \quad (1)$$

and $W(y_1, y_2, \dots, y_n) \neq 0$ for at least one point in I . Then every solution of (1) can be expressed as a linear continuation of the solutions $y_1, y_2, \dots, y_n$.

Linear Dependence and Independence

The functions $y_1, y_2, \dots, y_n$ are said to be linearly dependent in an interval I if there exists set of constants $c_1, c_2, \dots, c_n$ not all zero such that $c_1y_1 + c_2y_2 + \dots + c_ny_n = 0$ and $t \in I$

On the other hand, $y_1, y_2, \dots, y_n$ are said to be linearly independent in an interval I if for any set of constants $c_1, c_2, \dots, c_n$;

$$c_1y_1 + c_2y_2 + \dots + c_ny_n = 0$$

holds for all $c_i = 0, i = 1, 2, \dots, n$.

Example: Determine where the given set of functions is linearly dependent or independent

i. $f_1(t) = 2t - 3, f_2(t) = t^3 + 1,$

$f_3(t) = 2t^2 - t, f_4(t) = t^2 + t + 1$

ii. $f_1(t) = 2t - 3, f_2(t) = t^2 + 1,$

$f_3(t) = 2t^2 - t, f_4(t) = t^2 + t + 1$

Solution:

ii. The given functions are

$f_1(t) = 2t - 3, f_2(t) = t^2 + 1, f_3(t) = 2t^2 - t, f_4(t) = t^2 + t + 1$

Let c_1, c_2, c_3, c_4 be constants such that

$$\begin{aligned} c_1f_1 + c_2f_2 + c_3f_3 + c_4f_4 &= 0 \\ \Rightarrow c_1(2t - 3) + c_2(t^2 + 1) + c_3(2t^2 - t) + c_4(t^2 + t + 1) &= 0 \quad \dots \quad (1) \\ \Rightarrow (c_2 + 2c_3 + c_4)t^2 + (2c_1 - c_3 + c_4)t + (-3c_1 + c_2 + c_4) &= 0 \\ \Rightarrow c_2 + 2c_3 + c_4 &= 0 \quad \dots \quad (1) \\ 2c_1 - c_3 + c_4 &= 0 \quad \dots \quad (2) \\ \text{and } -3c_1 + c_2 + c_4 &= 0 \quad \dots \quad (3) \end{aligned}$$

Solving (1), (2) and (3), we get

$$c_1 = -\frac{2}{3}, c_2 = -\frac{13}{7}, c_3 = \frac{3}{7}, c_4 = 1.$$

With these values of c_1, c_2, c_3, c_4 (1) becomes

$$-\frac{2}{3}f_1 - \frac{13}{7}f_2 + \frac{3}{7}f_3 + f_4 = 0$$

$$\Rightarrow 2f_1 + 13f_2 - 3f_3 - 7f_4 = 0.$$

Hence,

f_1, f_2, f_3, f_4 are linearly dependent.

Theorem 3 Let $y_1, y_2, \dots, y_n$ be a fundamental set of solutions of

$$L[y] = y^{(n)} + p_1y^{(n-1)} + p_2y^{(n-2)} + \dots + p_{n-1}y' + p_ny = 0 \quad \dots (1)$$

on an interval I . Then, $y_1, y_2, \dots, y_n$ are linearly independent on I . Conversely, if $y_1, y_2, \dots, y_n$ are linearly independent solutions of (1) on I then they form fundamental set of solutions of (1).

Homogeneous Equations with Constant Coefficients

The n^{th} order linear homogeneous differential equation with constant coefficient is

$$L[y] = a_0y^{(n)} + a_1y^{(n-1)} + \dots + a_{n-1}y' + a_ny = 0 \quad \dots (1)$$

where $a_0, a_1, \dots, a_n$ are real constants

Let $y = e^{mt}$ be a solution of (1). Then,

$$L[e^{mt}] = e^{mt} [a_0m^n + a_1m^{n-1} + \dots + a_{n-1}m + a_n] = 0$$

Since $e^{mt} \neq 0$, then

$$a_0m^n + a_1m^{n-1} + \dots + a_{n-1}m + a_n = 0.$$

Then, $a_0m^n + a_1m^{n-1} + \dots + a_{n-1}m + a_n$ is called the Characteristic Polynomial and $a_0m^n + a_1m^{n-1} + \dots + a_{n-1}m + a_n = 0$ is called the Characteristic Equation/Auxiliary Equation (A.E.). Three different cases may arise.

Case I: Real and distinct roots

Case II: Repeated roots.

Case III: Complex roots

Example: Find the general solution of $2y''' - 4y'' - 2y' + 4y = 0$.

Solution

The given differential equation is

$$2y''' - 4y'' - 2y' + 4y = 0 \quad \dots (1)$$

The A.E. is

$$2m^3 - 4m^2 - 2m + 4 = 0$$

$$\text{or, } 2m^2(m - 2) - 2(m - 2) = 0$$

$$\text{or, } (m - 2)(2m^2 - 2) = 0$$

$$\text{or, } (m - 2)(m - 1)(m + 1) = 0$$

$$\therefore m = 1, -1, 2$$

Hence the required general solution is

$$y = c_1e^t + c_2e^{-t} + c_3e^{-2t}.$$

Example: Find the general solution of $y^{(6)} + y = 0$

Solution:

The given differential equation is $y^{(6)} + y = 0 \quad \dots (1)$

It's A.E. is

$$m^6 + 1 = 0$$

$$\Rightarrow m^6 = -1 = \cos\pi + i\sin\pi$$

$$\therefore m = \cos\frac{(\pi+2k\pi)}{6} + i\sin\frac{(\pi+2k\pi)}{6}, k = 0, 1, 2, 3, 4, 5.$$

$$\therefore m = \frac{\sqrt{3}}{2} - \frac{1}{2}i, \quad \frac{\sqrt{3}}{2} + \frac{1}{2}i, \quad i, \quad -i, \quad -\frac{\sqrt{3}}{2} + \frac{1}{2}i, \quad -\frac{\sqrt{3}}{2} - \frac{1}{2}i$$

Hence, the required general solution is

$$y = e^{\frac{\sqrt{3}}{2}t}(c_1\cos\frac{t}{2} + c_2\sin\frac{t}{2}) + e^{-\frac{\sqrt{3}}{2}t}(c_3\cos\frac{t}{2} + c_4\sin\frac{t}{2}) + (c_5\cos t + c_6\sin t)$$

Example: Find the general solution of $y^{(6)} - 3y^{(4)} + 3y'' - y = 0$.

Solution:

The given differential equation is

$$y^{(6)} - 3y^{(4)} + 3y'' - y = 0 \quad \dots (1).$$

It's A.E. is

$$m^6 - 3m^4 + 3m^2 - 1 = 0$$

$$m^6 - 3m^4 + 3m^2 - 1 = 0$$

$$\text{or, } M^3 - 3M^2 + 3M - 1 = 0$$

$$\text{or, } (M - 1)^3 = 0$$

$$\text{or, } (m^2 - 1)^3 = 0$$

$$\Rightarrow ((m - 1)(m + 1))^3 = 0$$

$$\Rightarrow (m - 1)^3(m + 1)^3 = 0$$

$\therefore m = 1$ (with multiplicity 3) and $m = -1$ (with multiplicity 3)

$\therefore$ The general solution is

$$y = (c_1e^t + c_2te^t + c_3t^2e^t) + (c_4e^{-t} + c_5te^{-t} + c_6t^2e^{-t})$$

Example: Find the solution of the IVP

$$4y''' + y' + 5y = 0,$$

$$y(0) = 2, \quad y'(0) = 1, \quad y''(0) = -1$$

How does the solution behave as $t \rightarrow \infty$.

Solution of Non-homogeneous Equations

$$L[y] = y^{(n)} + p_1 y^{(n-1)} + p_2 y^{(n-2)} + \dots + p_{n-1} y' + p_n y = g(t)$$

Method of Undetermined Coefficients

Method of undetermined coefficients for solving linear differential equations with constant coefficients

$$L[y] = a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = g(t).$$

The following table suggests the form of trial solution $Y(t)$ (for particular solution) to be used to a special form of $g(t)$.

Table

S.No.	Special Form of $g(t)$	Trial Solution
1	t^n OR $a_n t^n$ OR $a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0$	$A_n t^n + A_{n-1} t^{n-1} + \dots + A_1 t + A_0$
2	e^{at} OR pe^{at}	Ae^{at}
3	$p \cos at$ OR $q \sin at$ OR $p \cos at + q \sin at$	$A \sin at + B \cos at$
4	$a_n t^n e^{at}$ OR $(a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0) e^{at}$	$(A_n t^n + A_{n-1} t^{n-1} + \dots + A_1 t + A_0) e^{at}$
5	$pe^{bt} \sin at$ OR $qe^{bt} \cos at$ OR $(p \sin at + q \cos at) e^{bt}$	$(A \sin at + B \cos at) e^{bt}$
6	$t^n \sin at$ OR $a_n t^n \sin at$ OR $t^n \cos at$ OR $a_n t^n \cos at$ OR $(a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0) \sin at$ OR $(a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0) \cos at$	$(A_n t^n + A_{n-1} t^{n-1} + \dots + A_1 t + A_0) \sin at$ $+ (B_n t^n + B_{n-1} t^{n-1} + \dots + B_1 t + B_0) \cos at$

Note

1. If the RHS $g(t)$ of the given equation is a linear combination of more than one special forms of the above table, then the trial solution must be taken as the sum of the corresponding trial solution with appropriate constant coefficients to be determined later.
2. The trial solution indicated in the above table is modified in the following situations:

Situation A A term of $g(t)$ is also a term of the C.F. of the given equation

If a term of $g(t)$, u say, is also a term of the C.F. corresponding to a root with multiplicity m , then the trial solution $Y(t)$ of the table we introduce a term $x^m u$ plus the terms arising from it by differentiation.

For example, consider

$$(D - 2)^2(D + 3)y = e^{2t} + t^2$$

That is,

$$\frac{d^3y}{dt^3} - \frac{d^2y}{dt^2} - 8\frac{dy}{dt} + 12y = e^{2t} + t^2.$$

It's C.F. is $(c_1 + c_2t)e^{2t} + c_3e^{-3t}$.

Note that t^2 is not present in the C.F. So, the corresponding contribution to the trial solution $Y(t)$ is $A_2t^3 + A_1t + A_0$. Next, note carefully that e^{2t} is present in the C.F. to the root $m = 2$ with multiplicity 2. Hence, we add t^2e^{2t} and all the terms that arise from it by differentiating. Thus, the contribution to the trial solution $Y(t)$ corresponding to e^{2t} of $g(t)$ is taken as $B_2t^2e^{2t} + B_1te^{2t} + B_0e^{2t}$. Thus, the total P.S. is $Y(t) = A_2t^3 + A_1t + A_0 + B_2t^2e^{2t} + B_1te^{2t} + B_0e^{2t}$.

Situation B A term of $g(t)$ is $t^m u$ and u is a term of the C.F. of the given equation

If $t^r u$ is in $g(t)$ and u correspond to a root (with multiplicity m) of A.E. the trial solution $Y(t)$ must contain the term $t^{r+m}u$ plus terms arising from it by differentiation.

For example, consider

$$(D - 2)^3(D + 3)y = t^2e^{2t} + t^2$$

That is,

$$\frac{d^4y}{dt^4} - 3\frac{d^3y}{dt^3} - 6\frac{d^2y}{dt^2} + 28\frac{dy}{dt} - 24y = t^2e^{2t} + t^2.$$

It's C.F. is

$$(c_1 + c_2t + c_3t^2)e^{2t} + c_4e^{-3t}.$$

Note that t^2e^{2t} occurs in $g(t)$ and e^{2t} also present in C.F. corresponding to the root $m = 3$ with multiplicity 3. Hence, we employ $t^{2+3}u^{2t}$ i.e. t^5u^{2t} and all the terms that arise from it by derivative. Thus, the contribution to the trial solution $Y(t)$ corresponding to t^2e^{2t} of $g(t)$ is taken as

$$Y_1(t) = A_5t^5e^{2t} + A_4t^4e^{2t} + A_3t^3e^{2t} + A_2t^2e^{2t} + A_1te^{2t} + A_0.$$

Next, note carefully that the term t^2 in $g(t)$ is not present in the C.F. Thus, the contribution to the trial solution $Y(t)$ corresponding to t^2 of $g(t)$ is taken as $Y_2(t) = B_2t^2 + B_1t + B_0$. Thus, the total P.S. is $Y(t) = Y_1(t) + Y_2(t) = A_5t^5e^{2t} + A_4t^4e^{2t} + A_3t^3e^{2t} + A_2t^2e^{2t} + A_1te^{2t} + A_0 + B_2t^2 + B_1t + B_0$.

Examples

1. Solve by the method of undetermined coefficients: $\frac{d^2y}{dt^2} + 4y = t^2$.

Solution

The given equation is

$$\frac{d^2y}{dt^2} + 4y = t^2 \quad \dots \quad (1)$$

The A.E. is

$$m^2 + 4 = 0 \\ \Rightarrow m = \pm 2i$$

Therefore,

$$C.F. = c_1 \cos 2t + c_2 \sin 2t \quad \dots \quad (2)$$

where, c_1 and c_2 are arbitrary constants.

To find P.S.

Let a trial solution be

$$Y(t) = A_2t^2 + A_1t + A_0 \quad \dots \quad (3)$$

So,

$$Y'(t) = 2A_2t + A_1 \quad \dots \quad (4)$$

$$Y''(t) = 2A_2 \quad \dots \quad (5)$$

Using (3), (4) and (5) in (1), we get

$$2A_2 + 4(A_2t^2 + A_1t + A_0) = t^2 \\ 4A_2t^2 + 4A_1t + (2A_2 + 4A_0) = t^2 \quad \dots \quad (6)$$

Equating the coefficients of like terms, we get

$$\begin{aligned} 4A_2 &= 1, \quad 4A_1 = 0, \quad 2A_2 + 4A_0 = 0 \\ \Rightarrow A_2 &= \frac{1}{4}, \quad A_1 = 0, \quad 2 \times \frac{1}{4} + 4A_0 = 0 \\ \Rightarrow A_2 &= \frac{1}{4}, \quad A_1 = 0, \quad A_0 = -\frac{1}{8} \end{aligned}$$

Thus, the Particular solution is

$$Y(t) = \frac{1}{4}t^2 - \frac{1}{8}.$$

Hence, the required general solution is

$$\begin{aligned} y(t) &= C.F. + P.S. = c_1 \cos 2t + c_2 \sin 2t + \frac{t^2}{4} - \frac{1}{8}. \\ \therefore y(t) &= c_1 \cos 2t + c_2 \sin 2t + \frac{t^2}{4} - \frac{1}{8}. \end{aligned}$$

2. $y'' + 2y' + y = x - e^x$

Solution

The given equation is

$$y'' + 2y' + y = x - e^x \quad \dots \quad (1)$$

The A.E. is

$$\begin{aligned} m^2 + 2m + 1 &= 0 \\ \Rightarrow m &= -1 \text{ (multiplicity 2)} \end{aligned}$$

Therefore,

$$C.F. = (c_1 + c_2 x)e^{-x} \quad \dots \quad (2)$$

To find P.S.

Let the P.S. (trial solution) be

$$Y(t) = Ax + B + Ce^x$$

Therefore,

$$\begin{aligned} Y'(t) &= A + Ce^x \\ Y''(t) &= Ce^x \end{aligned}$$

Since $Y(t)$ is a solution of (1), we get

$$\begin{aligned} Ce^x + 2(A + Ce^x) + Ax + B + Ce^x &= x - e^x \\ \Rightarrow Ax + (2A + B) + 4Ce^x &= x - e^x \end{aligned}$$

Equating the coefficients of like terms, we get

$$\begin{aligned} A &= 1, \quad 2A + B = 0, \quad 4C = -1 \\ \Rightarrow A &= 1, \quad 2.1 + B = 0, \quad C = -\frac{1}{4} \\ \Rightarrow A &= 1, \quad B = -2, \quad C = -\frac{1}{4} \end{aligned}$$

Therefore,

$$Y(t) = x - 2 - \frac{1}{4}e^x$$

Hence, the general solution is

$$\begin{aligned} y(t) &= C.F. + P.S. \\ y(t) &= (c_1 + c_2 x)e^{-x} + x - 2 - \frac{1}{4}e^x \end{aligned}$$

3. $y'' + 3y' + 2y = x + \cos x$

Solution

The given equation is

$$y'' + 3y' + 2y = x + \cos x \quad \dots \quad (1)$$

The A.E. is

$$\begin{aligned} m^2 + 3m + 2 &= 0 \\ \Rightarrow (m + 1)(m + 2) &= 0 \\ \Rightarrow m &= -1, -2 \end{aligned}$$

Therefore,

$$C.F. = c_1 e^{-x} + c_2 e^{-2}$$

To find P.S.

Let the trial solution be

$$Y(t) = Ax + B + C \cos x + D \sin x$$

Then,

$$Y'(t) = A - C \sin x + D \cos x$$

$$Y''(t) = -C \cos x - D \sin x$$

Using these in (1) we get

$$-C \cos x - D \sin x + 3(A - C \sin x + D \cos x) + 2(Ax + B + C \cos x + D \sin x) = x + \cos x$$

$$\Rightarrow 2Ax + (-C + 3D + 2C) \cos x + (-D - 3C + 2D) \sin x + 3A + 2B = x + \cos x$$

$$\Rightarrow 2Ax + (3D + C) \cos x + (-3C + D) \sin x + 3A + 2B = x + \cos x$$

Equating the coefficients of like terms, we get

$$2A = 1, \quad 3D + C = 1, \quad -3C + D = 0, \quad 3A + 2B = 0$$

$$\Rightarrow A = \frac{1}{2}, \quad 3D + C = 1, \quad D = 3C, \quad 3 \cdot \frac{1}{2} + 2B = 0$$

$$\Rightarrow A = \frac{1}{2}, \quad 3D + C = 1, \quad D = 3C, \quad B = -\frac{3}{4}$$

$$\Rightarrow A = \frac{1}{2}, \quad 9C + C = 1, \quad D = 3C, \quad B = -\frac{3}{4}$$

$$\Rightarrow A = \frac{1}{2}, \quad C = \frac{1}{10}, \quad D = \frac{3}{10}, \quad B = -\frac{3}{4}$$

Therefore, the P.S. is

$$Y(t) = \frac{1}{2}x - \frac{3}{4} + \frac{1}{10} \cos x + \frac{3}{10} \sin x$$

And hence, the general solution is

$$y(t) = C.F. + P.S.$$

$$y(t) = c_1 e^{-x} + c_2 e^{-2} + \frac{1}{2}x - \frac{3}{4} + \frac{1}{10} \cos x + \frac{3}{10} \sin x$$

4. $y'' - 9y = t + e^{2t} - \sin 2t$

Solution

The given equation is

$$y'' - 9y = t + e^{2t} - \sin 2t \quad \dots \quad (1)$$

The A.E. is

$$m^2 - 9 = 0$$

$$\Rightarrow m = 3, -3$$

Therefore,

$$C.F. = c_1 e^{3t} + c_2 e^{-3t}$$

To find P.S.

Here, $g(t) = t + e^{2t} - \sin 2t$

Let the trial solution corresponding to the term t be $At + B$, corresponding to the term e^{2t} be Ce^{2t} and corresponding to the term $\sin 2t$ be $D \cos 2t + E \sin 2t$. Combining these together, the net trial solution becomes

$$Y(t) = At + B + Ce^{2t} + D \cos 2t + E \sin 2t$$

Therefore,

$$Y'(t) = A + 2Ce^{2t} - 2D \sin 2t + 2E \cos 2t$$

$$Y''(t) = 4Ce^{2t} - 4D \sin 2t - 4E \sin 2t$$

Putting these values in (1), we get

$$(4Ce^{2t} - 4D \sin 2t - 4E \sin 2t) - 9(At + B + Ce^{2t} + D \cos 2t + E \sin 2t) = t + e^{2t} - \sin 2t$$

$$\Rightarrow -9At - 5Ce^{2t} - 13D \cos 2t - 13E \sin 2t - 9B = t + e^{2t} - \sin 2t$$

Equating the coefficients of the like terms, we get

$$-9A = 1, \quad -5C = 1, \quad -13D = 0, \quad -13E = -1, \quad -9B = 0$$

$$\Rightarrow A = -\frac{1}{9}, \quad C = -\frac{1}{5}, \quad D = 0, \quad E = \frac{1}{13}, \quad B = 0$$

Therefore,

$$Y(t) = -\frac{1}{9}t - \frac{1}{5}e^{2t} + \frac{1}{13}\sin 2t$$

Hence, the general solution is

$$y(t) = C.F. + P.S.$$

$$y(t) = c_1 e^{3t} + c_2 e^{-3t} - \frac{1}{9}t - \frac{1}{5}e^{2t} + \frac{1}{13}\sin 2t$$

5. $y''' + 3y'' + 2y' = t^2 + 4t + 8$

Solution

The given equation is

$$y''' + 3y'' + 2y' = t^2 + 4t + 8 \quad \dots \quad (1)$$

The A.E. is

$$m^3 + 3m^2 + 2m = 0$$

$$\Rightarrow m(m+1)(m+2) = 0$$

$$\therefore m = 0, -1, -2$$

Therefore,

$$C.F. = c_1 + c_2 e^{-t} + c_3 e^{-2t}$$

To find P.S.

Here, $g(t) = t^2 + 4t + 8$

The constant term 8 in $g(t)$ and it occurs as constant c_1 in C.F. corresponding to the real root 0 with multiplicity 1. So, we choose a trial solution of (1) as

$$Y(t) = t(At^2 + Bt + C) = At^3 + Bt^2 + Ct$$

$$\Rightarrow Y'(t) = 3At^2 + 2Bt + C$$

$$\Rightarrow Y''(t) = 6At + 2B$$

$$\Rightarrow Y'''(t) = 6A$$

Using these in (1), we get

$$6A + 3(6At + 2B) + 2(3At^2 + 2Bt + C) = t^2 + 4t + 8$$

$$\Rightarrow 6At^2 + (18A + 4B)t + (6A + 6B + 2C) = t^2 + 4t + 8$$

Equating the coefficients of like terms, we get

$$6A = 1, \quad 18A + 4B = 4, \quad 6A + 6B + 2C = 8$$

$$\Rightarrow A = \frac{1}{6}, \quad 18 \times \frac{1}{6} + 4B = 4, \quad 6 \times \frac{1}{6} + 6B + 2C = 8$$

$$\Rightarrow A = \frac{1}{6}, \quad B = \frac{1}{4}, \quad 6 \times \frac{1}{6} + 6 \times \frac{1}{4} + 2C = 8$$

$$\Rightarrow A = \frac{1}{6}, \quad B = \frac{1}{4}, \quad C = \frac{11}{4}$$

Therefore,

$$Y(t) = t\left(\frac{1}{6}t^2 + \frac{1}{4}t + \frac{11}{4}\right) = \frac{1}{6}t^3 + \frac{1}{4}t^2 + \frac{11}{4}t$$

Hence, the general solution is

$$y(t) = C.F. + P.S.$$

$$y(t) = c_1 + c_2 e^{-t} + c_3 e^{-2t} + \frac{1}{6}t^3 + \frac{1}{4}t^2 + \frac{11}{4}t$$

6. $y''' + y' = 2t^2 + 4\sin t$

Solution

The given equation is

$$y''' + y' = 2t^2 + 4\sin t \quad \dots \quad (1)$$

The A.E. is

$$m^3 + m = 0$$

$$\Rightarrow m(m^2 + 1) = 0$$

$$\therefore m = 0, \quad i, \quad -i$$

Therefore,

$$C.F. = c_1 + c_2 \cos t + c_3 \sin t$$

To find P.S.

Here, $g(t) = 2t^2 + 4\sin t$.

The constant term c_1 in C.F. appears corresponding to the term $2t^2$ in $g(t)$ for the real root 0 with multiplicity one. So, the trial solution corresponding to $2t^2$ is taken as $t(At^2 + Bt + C)$. Also, $c_3 \sin t$ in C.F. corresponds to the term $4 \sin t$ in $g(t)$ corresponding to the complex root with multiplicity one. So, P.S. corresponding to $4 \sin t$ is taken as $t(D \cos t + E \sin t)$. Therefore, the net trial solution is

$$\begin{aligned} Y(t) &= t(At^2 + Bt + C) + t(D \cos t + E \sin t) = At^3 + Bt^2 + Ct + Dt \cos t + Et \sin t \\ \Rightarrow Y'(t) &= 3At^2 + 2Bt + C + D \cos t - Dts \sin t + E \sin t + Etc \cos t \\ \Rightarrow Y''(t) &= 6At + 2B - 2Ds \sin t - Dtc \cos t + 2E \cos t - Ets \sin t \\ \Rightarrow Y'''(t) &= 6A - 3Dc \cos t + Dts \sin t - 3E \sin t - Etc \cos t \end{aligned}$$

Substituting these values in (1), we get

$$\begin{aligned} (6A - 3Dc \cos t + Dts \sin t - 3E \sin t - Etc \cos t) \\ + (3At^2 + 2Bt + C + D \cos t - Dts \sin t + E \sin t + Etc \cos t) &= 2t^2 + 4 \sin t \\ \Rightarrow (6A + C) - 2Dc \cos t - 2E \sin t + 3At^2 + 2Bt &= 2t^2 + 4 \sin t \end{aligned}$$

Equating the coefficients of like terms, we get

$$\begin{aligned} (6A + C) &= 0, \quad -2D = 0, \quad -2E = 4, \quad 3A = 4, \quad 2B = 0 \\ \Rightarrow (6A + C) &= 0, \quad D = 0, \quad E = -2, \quad A = \frac{4}{3}, \quad B = 0 \\ \Rightarrow \left(6 \times \frac{4}{3} + C\right) &= 0, \quad D = 0, \quad E = -2, \quad A = \frac{4}{3}, \quad B = 0 \\ \Rightarrow C &= -8, \quad D = 0, \quad E = -2, \quad A = \frac{4}{3}, \quad B = 0 \end{aligned}$$

Therefore,

$$Y(t) = \frac{3}{4}t^3 - 8t - 2t \sin t$$

Hence, the required general solution is

$$y(t) = c_1 + c_2 \cos t + c_3 \sin t + \frac{3}{4}t^3 - 8t - 2t \sin t$$

$$7. y''' - 3y'' + 2y' = 3te^{2t} + 5t^2$$

Solution

The given equation is

$$y''' - 3y'' + 2y' = 3te^{2t} + 5t^2 \quad \dots \quad (1)$$

The A.E. is

$$\begin{aligned} m^3 - 3m^2 + 2m &= 0 \\ \Rightarrow m(m^2 - 3m + 2) &= 0 \\ \Rightarrow m(m - 1)(m - 2) &= 0 \\ \therefore m &= 0, 1, 2 \end{aligned}$$

Therefore,

$$C.F. = c_1 + c_2 e^t + c_3 e^{2t}$$

To find P.S.

Here, $g(t) = 3te^{2t} + 5t^2$

We see that te^{2t} is present in $g(t)$ where e^{2t} is a term of C.F. corresponding to the root 2 with multiplicity 1. So, the trial solution corresponding to it is taken as $t^{1+1}e^{2t} = t^2e^{2t}$ plus the terms that arise from it by differentiation. Thus, the contribution to the P.S. will be $At^2e^{2t} + Bte^{2t}$ which do not include the term Ce^{2t} because it is already in C.F. with arbitrary constant c_3 . Again, since the constant term c_1 appears in C.F. corresponding to the root 0 with multiplicity 1 in association with the term $5t^2$ in $g(t)$. So, the contribution of $5t^2$ in P.S. is taken as $t(Ct^2 + Dt + E)$.

So, the total trial solution is

$$\begin{aligned} Y(t) &= At^2e^{2t} + Bte^{2t} + Ct^3 + Dt^2 + Et \\ \Rightarrow Y'(t) &= 2Ate^{2t} + 2At^2e^{2t} + Be^{2t} + 2Bte^{2t} + 3Ct^2 + 2Dt + E \\ &= 2At^2e^{2t} + (2A + 2B)te^{2t} + Be^{2t} + 3Ct^2 + 2Dt + E \\ \Rightarrow Y''(t) &= 4Ate^{2t} + 4At^2e^{2t} + (2A + 2B)e^{2t} + (4A + 4B)te^{2t} + 2Be^{2t} + 6Ct + 2D \\ &= 4At^2e^{2t} + (6A + 4B)te^{2t} + (2A + 4B)e^{2t} + 6Ct + 2D \\ \Rightarrow Y'''(t) &= 8At^2e^{2t} + 8Ate^{2t} + (12A + 8B)te^{2t} + (6A + 4B)e^{2t} + (4A + 8B)e^{2t} + 6C \\ &= 8At^2e^{2t} + (20A + 8B)te^{2t} + (10A + 12B)e^{2t} + 6C \end{aligned}$$

Substituting these values in (1) we get

$$\begin{aligned}
 & y''' - 3y'' + 2y' = 3te^{2t} + 5t^2 \\
 & [8At^2e^{2t} + (20A + 8B)te^{2t} + (10A + 12B)e^{2t} + 6C] \\
 & - 3[4At^2e^{2t} + (6A + 4B)te^{2t} + (2A + 4B)e^{2t} + 6Ct + 2D] \\
 & + 2[2At^2e^{2t} + (2A + 2B)te^{2t} + Be^{2t} + 3Ct^2 + 2Dt + E] = 3te^{2t} + 5t^2 \\
 \Rightarrow & 6Ate^{2t} + (4A + B)e^{2t} + 3Ct^2 + (-18C + 4D)t + (6C - 6D + 2E) = 3te^{2t} + 5t^2
 \end{aligned}$$

Equating the coefficients of like terms, we get

$$\begin{aligned}
 6A &= 3, & 4A + B &= 0, & -18C + 4D &= 0, & 3C &= 5, & 6C - 6D + 2E &= 0 \\
 \Rightarrow A &= \frac{1}{2}, & 4 \times \frac{1}{2} + B &= 0, & -18C + 4D &= 0, & C &= \frac{5}{3}, & 6C - 6D + 2E &= 0 \\
 \Rightarrow A &= \frac{1}{2}, & B &= -2, & -18 \times \frac{5}{3} + 4D &= 0, & C &= \frac{5}{3}, & 6 \times \frac{5}{3} - 6D + 2E &= 0 \\
 \Rightarrow A &= \frac{1}{2}, & B &= -2, & D &= \frac{15}{2}, & C &= \frac{5}{3}, & 10 - 6 \times \frac{15}{2} + 2E &= 0 \\
 \Rightarrow A &= \frac{1}{2}, & B &= -2, & D &= \frac{15}{2}, & C &= \frac{5}{3}, & E &= \frac{35}{2}
 \end{aligned}$$

Therefore,

$$Y(t) = \frac{1}{2}t^2e^{2t} - 2te^{2t} + \frac{5}{3}t^3 + \frac{15}{2}t^2 + \frac{35}{2}t$$

Hence, the required general solution is

$$y(t) = c_1 + c_2e^t + c_3e^{2t} + \frac{1}{2}t^2e^{2t} - 2te^{2t} + \frac{5}{3}t^3 + \frac{15}{2}t^2 + \frac{35}{2}t$$

8. Find the general solution of $y''' - 3y'' + 3y' - y = 3e^t$

Solution

The given de is

$$y''' - 3y'' + 3y' - y = 3e^t \quad \dots \quad (1)$$

The A.E. is

$$\begin{aligned}
 m^3 - 3m^2 + 3m - 1 &= 0 \\
 \Rightarrow (m - 1)^3 &= 0 \\
 \therefore m &= 1, 1, 1.
 \end{aligned}$$

$\therefore$ Complementary Function is

$$y_c(t) = c_1e^t + c_2te^t + c_3t^2e^t = e^t(c_1 + c_2t + c_3t^2) \quad \dots \quad (2)$$

Let us assume the particular solution be

$y(t) = At^3e^t$. Then,

$$\begin{aligned}
 y'(t) &= Ae^t(3t^2 + t^3) \\
 y''(t) &= Ae^t(t^3 + 6t^2 + 6t) \\
 y'''(t) &= Ae^t(t^3 + 9t^2 + 18t + 6)
 \end{aligned}$$

Using these values in (1) we get

$$\begin{aligned}
 Ae^t(t^3 + 9t^2 + 18t + 6) - 3Ae^t(t^3 + 6t^2 + 6t) + 3Ae^t(3t^2 + t^3) - At^3e^t &= 3e^t \\
 \Rightarrow 6Ae^t &= 3e^t \Rightarrow A = \frac{1}{2}.
 \end{aligned}$$

$\therefore$ Particular solution is

$$Y(t) = \frac{1}{2}t^3e^t$$

Hence, the general solution is

$$y = C.F. + P.S. = e^t(c_1 + c_2t + c_3t^2) + \frac{1}{2}t^3e^t$$

9. Find the general solution of

$$y^{(4)} + 2y'' + y = 3\sin t - 5\cos t.$$

Solution

The given differential equation is

$$y(4) + 2y'' + y = 3\sin t - 5\cos t \quad \dots \quad (1)$$

It's A.E. is

$$m^4 + 2m^2 + 1 = 0$$

$$\text{or, } (m^2 + 1)^2 = 0$$

$$(m^2 + 1)(m^2 + 1) = 0$$

$$\therefore m = i, i, -i, -i$$

$\therefore$ the complementary solution is

$$y_c(t) = c_1 \cos t + c_2 \sin t + c_3 t \cos t + c_4 t \sin t$$

Let the particular solution of (1) be

$$Y(t) = At^2 \sin t + Bt^2 \cos t. \text{ Then,}$$

$$Y'(t) = 2At \sin t + At^2 \cos t + 2Bt \cos t - Bt^2 \sin t$$

$$Y''(t) = 2A \sin t + 4At \cos t - At^2 \sin t + 2B \cos t - 3Bt \sin t - Bt^2 \cos t$$

$$Y'''(t) = 6A \cos t - 6At \sin t - At^2 \cos t - 5B \sin t - 5Bt \cos t + Bt^2 \sin t$$

and

$$Y^{(4)}(t) = -12A \sin t + 4At \cos t + At^2 \sin t - 10B \cos t + 5B \sin t - 7Bt \sin t + Bt^2 \cos t$$

Using these values in (1) we get

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$-8A \sin t - 8B \cos t = 3 \sin t - 5 \cos t.$$

Equating the coefficients of like terms, we get

$$-8A = 3 \Rightarrow A = -\frac{3}{8} \text{ and } -8B = -5 \Rightarrow B = \frac{5}{8}$$

$\therefore$ the particular solution is

$$Y(t) = -\frac{3}{8}t^2 \sin t + \frac{5}{8}t^2 \cos t$$

Hence, the required general solution is

$$y(t) = y_c(t) + Y(t)$$

$$\text{or, } y(t) = c_1 \cos t + c_2 \sin t + c_3 t \cos t + c_4 t \sin t - \frac{3}{8}t^2 \sin t + \frac{5}{8}t^2 \cos t$$

10. Find the general solution of

$$y''' + y'' + y' + y = e^{-t} + 4t$$

Solution

The given differential equation is

$$y''' + y'' + y' + y = e^{-t} + 4t \quad \dots \quad (1)$$

The A.E. of (1) is

$$m^3 + m^2 + m + 1 = 0$$

$$\Rightarrow (m^2 + 1)(m + 1) = 0$$

$$\therefore m = -1, i, -i$$

$$\therefore y_c(t) = c_1 e^{-t} + c_2 \cos t + c_3 \sin t.$$

From (1), we see that the P.S. of (1) is the sum of the P.S. of

$$y''' + y'' + y' + y = e^{-t} \quad \dots \quad (2)$$

and

$$y''' + y'' + y' + y = 4t \quad \dots \quad (3)$$

Let $Y_1(t) = Ate^{-t}$ be a P.S. of (2). Then,

$$\begin{aligned} Y_1'(t) &= Ae^{-t} - Ate^{-t} \\ Y_1''(t) &= -Ae^{-t} - Ae^{-t} + Ate^{-t} = -2Ae^{-t} + Ate^{-t} \\ Y_1'''(t) &= 2Ae^{-t} + Ae^{-t} - Ate^{-t} = 3Ae^{-t} - Ate^{-t} \end{aligned}$$

Using these values in (2) we get

$$\begin{aligned} 3Ae^{-t} - Ate^{-t} - 2Ae^{-t} + Ate^{-t} + Ae^{-t} - Ate^{-t} + Ate^{-t} &= e^{-t} \\ \Rightarrow 2Ae^{-t} &= e^{-t} \Rightarrow A = \frac{1}{2} \end{aligned}$$

$$Y_1(t) = \frac{1}{2}te^{-t}.$$

Next, let $Y_2(t) = At + B$ be a P.S. of (3).

$$\therefore Y_2'(t) = A, Y_2''(t) = 0 \text{ and } Y_2'''(t) = 0$$

Using these in (3), we get

$$\begin{aligned} 0 + 0 + A + At + B &= 4t \\ \Rightarrow At + A + B &= 4t \\ \Rightarrow A = 4, \text{ and } A + B &= 0 \\ \therefore A = 4, \quad B &= -4 \end{aligned}$$

$\therefore$ P.S. of (3) is

$$Y_2(t) = 4t - 4$$

So, P.S. of (1) is

$$Y(t) = Y_1(t) + Y_2(t) = \frac{1}{2}te^{-t} + 4(t - 1).$$

Hence, the general solution is

$$y = c_1e^{-t} + c_2\cos t + c_3\sin t + \frac{1}{2}te^{-t} + 4(t - 1).$$

11. $y''' + 3y'' + 3y' + y = e^{-t}(t + 5)$

Solution

The given equation is

$$y''' + 3y'' + 3y' + y = e^{-t}(t + 5) \quad \dots \quad (1)$$

The A.E. is

$$\begin{aligned} m^3 + 3m^2 + 3m + 1 &= 0 \\ \Rightarrow (m + 1)^3 &= 0 \\ \therefore m &= -1, -1, -1 \end{aligned}$$

Therefore,

$$C.F. = (c_1 + c_2t + c_3t^2)e^{-t} \quad \dots \quad (2)$$

where, c_1, c_2, c_3 are arbitrary constants.

To find C.F.

Here, $g(t) = e^{-t}(t + 5)$.

We see that the term e^{-t} of $g(t)$ is a term of C.F. corresponding to the root -1 with multiplicity 3. So, the trial solution will contain the term $t^{1+3}e^{-t} = t^4e^{-t}$ plus the terms arising from it by differentiation. Therefore, let the trial solution be

$$Y(t) = At^4e^{-t} + Bt^3e^{-t} = (At^4 + Bt^3)e^{-t}$$

In which the terms Ct^2e^{-t} , Dte^{-t} , Ee^{-t} are not included because they already appear in C.F. with arbitrary constants c_1, c_2, c_3 .

Proceeding as before, we get the general solution as

$$y(t) = (c_1 + c_2t + c_3t^2)e^{-t} + (At^4 + Bt^3)e^{-t}.$$

The Variation of Parameters

Let

$$L[y] = y^{(n)} + p_1y^{(n-1)} + \dots + p_{n-1}y' + p_ny = g(t) \quad \dots \quad (1)$$

be a non-homogeneous n^{th} order linear differential equation.

Let $y_1, y_2, \dots, y_n$ be a set of fundamental solutions of the corresponding homogeneous equation

$$L[y] = 0.$$

Then, the complementary solution of (1) is

$$y_c(t) = c_1 y_1(t) + c_2 y_2(t) + \dots + c_n y_n(t) \quad \dots \quad (2)$$

Let the particular solution of (1) be

$$Y(t) = u_1(t)y_1(t) + u_2(t)y_2(t) + \dots + u_n(t)y_n(t) \quad \dots \quad (3)$$

Then, differernting (3) we get

$$Y'(t) = (u_1 y_1' + u_2 y_2' + \dots + u_n y_n') + (u_1' y_1 + u_2' y_2 + \dots + u_n' y_n) \quad \dots \quad (4)$$

Assume

$$u_1' y_1 + u_2' y_2 + \dots + u_n' y_n = 0 \quad \dots \quad (5)$$

So (4) reduces to

$$Y'(t) = u_1 y_1' + u_2 y_2' + \dots + u_n y_n' \quad \dots \quad (6)$$

Continuing the above process in the similar manner upto $n - 1$ steps we get

$$Y^{(m)}(t) = u_1 y_1^{(m)} + u_2 y_2^{(m)} + \dots + u_n y_n^{(m)} \quad \dots \quad (7)$$

where $m = 0, 1, 2, \dots, n - 1$, with the following $n - 1$ conditions

$$u_1' y_1^{(m-1)} + u_2' y_2^{(m-1)} + \dots + u_n' y_n^{(m-1)} = 0 \quad \dots \quad (8)$$

$$m = 1, 2, 3, \dots, n - 1$$

Also, the n^{th} derivative of $Y(t)$ is

$$Y^{(n)}(t) = (u_1 y_1^{(n)} + u_2 y_2^{(n)} + \dots + u_n y_n^{(n)}) + (u_1' y_1^{(n-1)} + u_2' y_2^{(n-1)} + \dots + u_n' y_n^{(n-1)}) + \dots + \dots \quad \dots \quad (9)$$

Substituting the values of $Y, Y', Y'', \dots, Y^{(n)}$ in equation (1), we get

$$\begin{aligned} & \left[(u_1 y_1^{(n)} + \dots + u_n y_n^{(n)}) + (u_1' y_1^{(n-1)} + \dots + u_n' y_n^{(n-1)}) \right] + p_1(t) [u_1 y_1^{(n-1)} + \dots + u_n y_n^{(n-1)}] \\ & + p_2(t) [u_1 y_1^{(n-2)} + \dots + u_n y_n^{(n-2)}] + \dots + p_{n-1}(t) [u_1 y_1' + \dots + u_n y_n'] \\ & + p_n(t) [u_1 y_1 + u_2 y_2 + \dots + u_n y_n] = g(t) \end{aligned}$$

Or,

$$\begin{aligned} & \left[u_1' y_1^{(n-1)} + \dots + u_n' y_n^{(n-1)} \right] + u_1 \left[y_1^{(n)} + p_1 y_1^{(n-1)} + p_2 y_1^{(n-2)} + \dots + p_n y_1 \right] \\ & + u_2 \left[y_2^{(n)} + p_1 y_2^{(n-1)} + p_2 y_2^{(n-2)} \dots + p_n y_2 \right] \\ & + u_3 \left[y_3^{(n)} + p_1 y_3^{(n-1)} + p_2 y_3^{(n-2)} \dots + p_n y_3 \right] + \dots \\ & + u_n \left[y_n^{(n)} + p_1 y_n^{(n-1)} + p_2 y_n^{(n-2)} \dots + p_n y_n \right] = g(t) \end{aligned}$$

Since, $L[y_i] = 0$, for all $i = 1, 2, \dots, n$ in, we get

$$u_1' y_1^{(n-1)} + \dots + u_n' y_n^{(n-1)} = g(t) \quad \dots \quad (10)$$

From equation (8) and (10), we get a system of non-homogeneous linear equation

$$\begin{aligned} & y_1 u_1' + y_2 u_2' + \dots + y_n u_n' = 0 \\ & y_1' u_1' + y_2' u_2' + \dots + y_n' u_n' = 0 \\ & y_1'' u_1' + y_2'' u_2' + \dots + y_n'' u_n' = 0 \\ & \dots \dots \dots \dots \dots \dots \dots \\ & \dots \dots \dots \dots \dots \dots \dots \end{aligned}$$

$$y_1^{(n-1)} u_1' + y_2^{(n-1)} u_2' + \dots + y_n^{(n-1)} u_n' = g(t)$$

with $u_1', u_2', \dots, u_n'$ as unknown variables. Using Cramer's Rule, the solution of the above system is

$$u_m'(t) = \frac{g(t)W_m(t)}{W(t)}, m = 1, 2, \dots, n \quad \dots \quad (11)$$

where, $W(t) = \begin{vmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & y_2' & \dots & y_n' \\ \dots & \dots & \dots & \dots \\ y_1^{(n-1)} & y_2^{(n-1)} & \dots & y_n^{(n-1)} \end{vmatrix}$ and $W_m(t)$ is a determinant obtained by replacing the m th. Column of $W(t)$ by the column $\begin{bmatrix} 0 \\ 0 \\ \dots \\ 1 \end{bmatrix}$.

Integrating equation (11), we get

$$u_m(t) = \int_{t_0}^t \frac{g(s)W_m(s)}{W(s)} ds, \quad m = 1, 2, \dots, n$$

for some $t_0 \in I$, where $p_1, p_2, \dots, p_n, g(t)$ are continuous in I .

Hence, $Y(t) = y_1 u_1 + y_2 u_2 + \dots + y_n u_n = \sum_{m=1}^n y_m u_m$.

Example Use the method of variation of parameter to determine the general solution of the differential equation $y''' - 2y'' - y' + 2y = e^{4t}$

Solution:

The given differential equation is

$$y''' - 2y'' - y' + 2y = e^{4t} \quad \dots \quad (1)$$

Here, $g(t) = e^{4t}$.

The A.E. of (1) is

$$\begin{aligned} m^3 - 2m^2 - m + 2 &= 0 \\ (m-2)(m-1)(m+1) &= 0 \\ \therefore m &= -1, 1, 2 \end{aligned}$$

$\therefore$ the complementary solution is

$$y_c(t) = c_1 e^{-t} + c_2 e^t + c_3 e^{2t}$$

Suppose $y_1(t) = e^{-t}, y_2(t) = e^t$ and $y_3(t) = e^{2t}$

Therefore, $y_1'(t) = -e^{-t}, y_2'(t) = e^t$ and $y_3'(t) = 2e^{2t}$,

and $y_1''(t) = e^{-t}, y_2''(t) = e^t$ and $y_3''(t) = 4e^{2t}$.

$$\begin{aligned} \therefore W(t) &= \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-t} & e^t & e^{2t} \\ -e^{-t} & e^t & 2e^{2t} \\ e^{-t} & e^t & 4e^{2t} \end{vmatrix} \\ &= e^{2t} \begin{vmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \\ 1 & 1 & 4 \end{vmatrix} = e^{2t} \begin{vmatrix} 0 & 1 & 1 \\ -2 & 1 & 2 \\ 0 & 1 & 4 \end{vmatrix} = e^{2t} \times 2(4-1) = 6e^{2t} \neq 0. \end{aligned}$$

$\therefore y_1, y_2, y_3$ form a fundamental set of solutions of corresponding homogeneous equation of (1).

Now,

$$\begin{aligned} W_1(t) &= \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & e^t & e^{2t} \\ 0 & e^t & 2e^{2t} \\ 1 & e^t & 4e^{2t} \end{vmatrix} = e^{3t} \\ W_2(t) &= \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-t} & 0 & e^{2t} \\ -e^{-t} & 0 & 2e^{2t} \\ e^{-t} & 1 & 4e^{2t} \end{vmatrix} = -3e^t \\ W_3(t) &= \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^{-t} & e^t & 0 \\ -e^{-t} & e^t & 0 \\ e^{-t} & e^t & 1 \end{vmatrix} = 2. \end{aligned}$$

$$u_1(t) = \int \frac{g(t)W_1(t)}{W(t)} dt = \int \frac{e^{4t} \times e^{3t}}{6e^{2t}} dt = \frac{1}{6} \int e^{5t} dt = \frac{e^{5t}}{30}$$

$$u_2(t) = \int \frac{g(t)W_2(t)}{W(t)} dt = \int \frac{e^{4t} \times -3e^t}{6e^{2t}} dt = -\frac{1}{2} \int e^{3t} dt = -\frac{e^{3t}}{6}$$

and

$$u_3(t) = \int \frac{g(t)W_3(t)}{W(t)} dt = \int \frac{e^{4t} \times 2}{6e^{2t}} dt = \frac{1}{3} \int e^{2t} dt = \frac{e^{2t}}{6}$$

Thus, the particular solution is

$$Y(t) = y_1(t)u_1(t) + y_2(t)u_2(t) + y_3(t)u_3(t)$$

$$= e^{-t} \times \frac{e^{5t}}{30} + e^t \times \left(-\frac{e^{3t}}{6}\right) + e^{2t} \times \frac{e^{2t}}{6} = \frac{e^{4t}}{30}.$$

Hence, the general solution of (1) is

$$y(t) = y_c(t) + Y(t) = c_1 e^{-t} + c_2 e^t + c_3 e^{2t} + \frac{e^{4t}}{30}$$

Example Use the method of variation of parameter to determine the general solution of the differential equation $y''' - y'' + y' - y = e^{-t} \sin t$

Solution

The given differential equation is

$$y''' - y'' + y' - y = e^{-t} \sin t \quad \dots \quad (1)$$

Here, $g(t) = e^{-t} \sin t$.

The A.E. of (1) is

$$m^3 - m^2 + m - 1 = 0$$

$$\rightarrow m^2(m - 1) + (m - 1) = 0$$

$$\rightarrow (m - 1)(m^2 + 1) = 0$$

$$\rightarrow m = 1, i, -i$$

$\therefore$ the complementary solution is

$$y_c(t) = c_1 e^t + c_2 \cos t + c_3 \sin t.$$

Suppose $y_1(t) = e^t, y_2(t) = \cos t$ and $y_3(t) = \sin t$

Therefore, $y_1'(t) = e^t, y_2'(t) = -\sin t$ and $y_3'(t) = \cos t$

and $y_1''(t) = e^t, y_2''(t) = -\cos t$ and $y_3''(t) = -\sin t$.

$$\therefore W(t) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^t & \cos t & \sin t \\ e^t & -\sin t & \cos t \\ e^t & -\cos t & -\sin t \end{vmatrix}$$

$$= e^t \begin{vmatrix} 1 & \cos t & \sin t \\ 1 & -\sin t & \cos t \\ 1 & -\cos t & -\sin t \end{vmatrix} = e^t \begin{vmatrix} 1 & \cos t & \sin t \\ 0 & -\sin t - \cos t & \cos t - \sin t \\ 0 & -2\cos t & -2\sin t \end{vmatrix} = -2e^t \times$$

$$\begin{vmatrix} -\sin t - \cos t & -\sin t + \cos t \\ \cos t & \sin t \end{vmatrix} = -2e^t \times (-\sin^2 t - \sin t \cos t + \sin t \cos t - \cos^2 t) = -2e^t \times$$

$$-1 = 2e^t \neq 0.$$

$\therefore y_1, y_2, y_3$ form a fundamental set of solutions of corresponding homogeneous equation of (1).

Now,

$$W_1(t) = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ 1 & -\cos t & -\sin t \end{vmatrix} = 1$$

$$W_2(t) = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^t & 0 & \sin t \\ e^t & 0 & \cos t \\ e^t & 1 & -\sin t \end{vmatrix} = e^t (\sin t - \cos t)$$

$$W_3(t) = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^t & \cos t & 0 \\ e^t & -\sin t & 0 \\ e^t & -\cos t & 1 \end{vmatrix} = -e^t(\sin t + \cos t).$$

$$\text{Again, } u_1(t) = \int \frac{g(t)W_1(t)}{W(t)} dt = \int \frac{e^{-t}\sin t \times 1}{2e^t} dt = \frac{1}{2} \int e^{-2t} \sin t dt = \frac{e^{-2t}(-2\sin t - \cos t)}{2(-2)^2 + 1} = \frac{-2\sin t - \cos t}{10e^{2t}}$$

$$u_2(t) = \int \frac{g(t)W_2(t)}{W(t)} dt = \int \frac{e^{-t}\sin t \times e^t(\sin t - \cos t)}{2e^t} dt = \frac{1}{2} \int e^{-t}(\sin^2 t - \sin t \cos t) dt$$

$$= -\frac{e^{3t}}{6}$$

and

$$u_3(t) = \int \frac{g(t)W_3(t)}{W(t)} dt = \int \frac{e^{4t} \times 2}{6e^{2t}} dt = \frac{1}{3} \int e^{2t} dt = \frac{e^{2t}}{6}$$

.....

Example Given that $y_1(t) = e^t, y_2(t) = te^t$ and $y_3(t) = e^{-t}$ are the solutions of the homogeneous equation corresponding to $y''' - y'' - y' + y = g(t)$, determine a particular solution.

Solution

The given differential equation is

$$y''' - y'' - y' + y = g(t) \quad \dots \quad (1)$$

Also,

$$y_1(t) = e^t, y_2(t) = te^t \text{ and } y_3(t) = e^{-t}$$

are solutions of the homogeneous equation

$$y''' - y'' - y' + y = 0 \quad \dots \quad (2)$$

Now,

$$y_1'(t) = e^t, \quad y_2'(t) = e^t + te^t, \quad y_3'(t) = -e^{-t}$$

$$y_1''(t) = e^t, \quad y_2''(t) = 2e^t + te^t, \quad y_3''(t) = e^{-t}$$

The Wronskian is

$$W(t) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^t & te^t & e^{-t} \\ e^t & e^t + te^t & -e^{-t} \\ e^t & 2e^t + te^t & e^{-t} \end{vmatrix}$$

$$= e^t \cdot e^{-t} \begin{vmatrix} 1 & te^t & 1 \\ 1 & e^t + te^t & -1 \\ 1 & 2e^t + te^t & 1 \end{vmatrix} = \begin{vmatrix} 0 & te^t & 1 \\ 2 & e^t + te^t & -1 \\ 0 & 2e^t + te^t & 1 \end{vmatrix}$$

$$= -2 \begin{vmatrix} te^t & 1 \\ 2e^t + te^t & 1 \end{vmatrix} = -2 \begin{vmatrix} -2e^t & 0 \\ 2e^t + te^t & 1 \end{vmatrix} = 4e^t \neq 0.$$

Therefore, y_1, y_2 and y_3 form a fundamental set of solution of (2).

Now,

$$W_1(t) = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & te^t & e^{-t} \\ 0 & e^t + te^t & -e^{-t} \\ 1 & 2e^t + te^t & e^{-t} \end{vmatrix} = \begin{vmatrix} te^t & e^{-t} \\ e^t + te^t & -e^{-t} \end{vmatrix}$$

$$= \begin{vmatrix} e^t + 2te^t & 0 \\ e^t + te^t & -e^{-t} \end{vmatrix} = -1 - 2t$$

$$W_2(t) = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^t & 0 & e^{-t} \\ e^t & 0 & -e^{-t} \\ e^t & 1 & e^{-t} \end{vmatrix} = 2$$

and

$$W_3(t) = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^t & te^t & 0 \\ e^t & e^t + te^t & 0 \\ e^t & 2e^t + te^t & 1 \end{vmatrix} = e^{2t}$$

Next,

$$\begin{aligned} u_1(t) &= \int_{t_0}^t \frac{g(s)W_1(s)}{W(s)} ds = \int_{t_0}^t \frac{g(s)(-1-2s)}{4e^s} ds \\ &= -\frac{1}{4} \int_{t_0}^t e^{-s} g(s)(1+2s) ds \\ u_2(t) &= \int_{t_0}^t \frac{g(s)W_2(s)}{W(s)} ds = \int_{t_0}^t \frac{g(s)(2)}{4e^s} ds \\ &= \frac{1}{2} \int_{t_0}^t g(s)e^{-s} ds \\ u_3(t) &= \int_{t_0}^t \frac{g(s)W_3(s)}{W(s)} ds = \int_{t_0}^t \frac{g(s)(e^{2s})}{4e^s} ds \\ &= \frac{1}{4} \int_{t_0}^t g(s)e^s ds \end{aligned}$$

Substituting these values, the particular solution becomes

$$\begin{aligned} Y(t) &= y_1(t)u_1(t) + y_2(t)u_2(t) + y_3(t)u_3(t) \\ &= e^t \times -\frac{1}{4} \int_{t_0}^t e^{-s} g(s)(1+2s) ds + te^t \times \frac{1}{2} \int_{t_0}^t g(s)e^{-s} ds + e^{-t} \times \frac{1}{4} \int_{t_0}^t g(s)e^s ds \\ &= \frac{1}{4} \int_{t_0}^t [e^{t-s}\{-1+2(t-s)\} + e^{-(t-s)}] g(s) ds \end{aligned}$$

Example Find the general solution of the differential equation $y''' + y' = \sec t$, $-\frac{\pi}{2} < t < \frac{\pi}{2}$, by the method of variation of parameter. Also, solve the initial value problem for this equation with the initial conditions $y(0) = 2, y'(0) = 1, y''(0) = -2$.

Solution

The given differential equation is

$$y''' + y' = \sec t, -\frac{\pi}{2} < t < \frac{\pi}{2} \quad \dots \quad (1)$$

Also, the initial conditions are

$$y(0) = 2, y'(0) = 1, y''(0) = -2 \quad \dots \quad (2)$$

The Auxiliary Equation of (1) is

$$m^3 + m = 0 \rightarrow m(m^2 + 1) = 0.$$

Solving this equation, we get

$$m = 0, i, -i$$

Therefore, the general solution of homogeneous equation corresponding to (1) is

$$y_c(t) = c_1 + c_2 \cos t + c_3 \sin t \quad \dots \quad (3)$$

Let

$$y_1(t) = 1, y_2(t) = \cos t, y_3(t) = \sin t$$

Therefore,

$$\begin{aligned} y_1'(t) &= 0, y_2'(t) = -\sin t, y_3'(t) = \cos t \\ y_1''(t) &= 0, y_2''(t) = -\cos t, y_3''(t) = -\sin t \end{aligned}$$

Now,

$$W(t) = \begin{vmatrix} 1 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ 0 & -\cos t & -\sin t \end{vmatrix} = 1 \neq 0.$$

Therefore, $y_1(t) = 1, y_2(t) = \cos t, y_3(t) = \sin t$ form a fundamental set of solution of the homogeneous equation corresponding to (1)

And

$$W_1(t) = \begin{vmatrix} 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ 1 & -\cos t & -\sin t \end{vmatrix} = 1, W_2(t) = \begin{vmatrix} 1 & 0 & \sin t \\ 0 & 0 & \cos t \\ 0 & 0 & -\sin t \end{vmatrix} = \cos t,$$

$$W_3(t) = \begin{vmatrix} 1 & \cos t & 0 \\ 0 & -\sin t & 0 \\ 0 & -\cos t & 1 \end{vmatrix} = -\sin t.$$

Therefore,

$$u_1'(t) = \frac{g(t)W_1(t)}{W(t)}$$

$$\Rightarrow u_1(t) = \int \frac{g(t)W_1(t)}{W(t)} dt = \int \frac{\sec t \times 1}{1} dt = \int \sec t dt = \ln |\sec t + \tan t|$$

And similarly,

$$u_2(t) = \int \frac{g(t)W_2(t)}{W(t)} dt = \int \frac{\sec t \times \cos t}{1} dt = \int dt = t$$

$$u_3(t) = \int \frac{g(t)W_3(t)}{W(t)} dt = \int \frac{\sec t \times -\sin t}{1} dt = -\int \tan t dt = \ln |\cos t|$$

Therefore, the particular solution is

$$Y(t) = y_1 u_1 + y_2 u_2 + y_3 u_3$$

$$= 1 \cdot \ln |\sec t + \tan t| + \cos t \times t + \sin t \times \ln |\cos t|$$

$$\therefore Y(t) = \ln |\sec t + \tan t| + t \cos t + \sin t \ln |\cos t|.$$

Thus, the general solution of (1) is

$$y(t) = y_c(t) + Y(t)$$

$$\Rightarrow y(t) = c_1 + c_2 \cos t + c_3 \sin t + \ln |\sec t + \tan t| + t \cos t + \sin t \ln |\cos t|$$

Also,

$$y'(t) = -c_2 \sin t + c_3 \cos t + \sec t + \cos t - t \sin t - \tan t \sin t + \cos t \ln |\cos t|$$

$$y''(t) = -c_2 \cos t - c_3 \sin t + \sec t \tan t - \sin t - \sin t - t \cos t - \sec^2 t \sin t - \tan t \cos t$$

$$- \sin t \log \cos t - \sin t$$

Using the initial conditions (2), we get

$$2 = c_1 + c_2$$

$$1 = c_3 + 1 + 1 \rightarrow c_3 = 1$$

$$-2 = -c_2 \rightarrow c_2 = 2$$

Using these values in first equation, we get $c_1 = 0$. Now, substituting these values of c_1, c_2, c_3 , we get the required solution of the IVP as

$$y(t) = 2 \cos t + \sin t + \ln |\sec t + \tan t| + t \cos t + \sin t \ln |\cos t|$$